

SCOTTIE D. MURRELL

SOFTWARE / XR ENGINEER | FULL-STACK, AI & REAL-TIME SYSTEMS

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PROFESSIONAL SUMMARY

Software engineer and technical lead delivering full-stack platforms, real-time multiplayer systems, XR/AR training applications, AI-enabled simulations, and Linux-based services. Led multidisciplinary development teams of approximately 15–20 contributors and translated partner requirements into deployable software for healthcare training, workforce development, professional networking, and immersive systems. Hands-on across frontend engineering, APIs, databases, networking, live-media integration, 3D/XR development, containerization, debugging, and deployment.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, C#, Python, Java, JavaScript, TypeScript, Swift, MATLAB
- **Full-Stack / Web:** React.js, Angular, Next.js, HTML5, CSS3, WebGL, Three.js, BabylonJS, Node.js, Express.js, MERN, MEAN, .NET
- **Backend / Data / Real-Time:** REST APIs, WebSockets, Microservices, MVC, Authentication, MongoDB, MySQL, SQL, API Integration
- **XR / Interactive / Game:** Unity, Babylon.js, WebXR, OpenXR, Vuforia, VR Simulation Development, AR Applications, Interactive 3D
- **Systems / DevOps / Cloud:** Linux, Unix, Ubuntu, Bash/Shell, TCP/IP, Server Configuration, Docker, Docker Compose, AWS, GitHub, GitLab, Jenkins, CI/CD
- **AI & Developer Tooling:** Claude.ai, OpenAI Codex, GitHub Copilot, Cursor, OpenAI APIs, LLMs, RAG, MCP, Visual Studio, VS Code, Xcode

SELECTED ENGINEERING PROJECTS

- **Da Vinci Surgical Robot VR Training Integration:** Built a real-time systems-integration pipeline that routed a live camera feed from a Da Vinci surgical-robot training environment into VR for clinician training. Integrated capture hardware, live-media routing, Unity-based visualization, and stereo rendering while troubleshooting latency and hardware/software compatibility. **Technologies:** *Unity, C#, VR, live video, capture hardware, OBS, custom shader / stereo rendering*
- **Multiplayer Game & Simulation Server Architecture:** Designed and implemented full-stack server architectures for multiplayer games and simulations, supporting client/server communication, shared application state, real-time interactions, backend services, and Linux-based deployment. **Technologies:** *Node.js, Express.js, WebSockets, REST APIs, Linux, Docker, networking*
- **Industry Workforce & Training Platform (EquipmentShare-related):** Developed database-backed full-stack software for company information access, workforce interaction, and training workflows. Worked across interface development, backend services, data integration, authentication, testing, and deployment-oriented architecture. **Technologies:** *React/Angular, Node.js, Express.js, REST, databases, authentication, Docker*
- **AI-Guided Laboratory Simulation:** Developed interactive laboratory simulations that use AI/LLM-based guidance to instruct users through equipment, procedures, process steps, and training interactions, combining application logic with conversational assistance. **Technologies:** *LLMs, OpenAI APIs, RAG/MCP concepts, interactive simulation, full-stack integration*
- **Professional Networking Platform:** Created a LinkedIn-style full-stack social platform connecting faculty and graduate students around research, employment, collaboration, profiles, and professional opportunities. **Technologies:** *Full-stack JavaScript/TypeScript, REST APIs, authentication, database-backed application design*
- **VR Health & Body Image Training:** Led development of an immersive training application addressing health, body awareness, and body image. Coordinated interaction logic, UI/UX, content integration, hardware deployment, testing, and iterative refinement as part of a multidisciplinary XR development program. **Technologies:** *Unity, C#, VR/XR, interaction systems, UI/UX, hardware deployment*
- **MeteorologyXR Mobile AR Application (iOS & Android):** Developed and published a cross-platform augmented-reality application for iPhone and Android, released through the Apple App Store and Google Play. Built the application in Unity with Vuforia image-target recognition and used Android Studio and Xcode for platform-specific builds and mobile deployment. Students could scan target images during class to view 3D cloud formations while accessing supporting audio and instructional text as the instructor explained cloud formation. **Technologies:** *Unity / Vuforia / Android Studio / Xcode / iOS / Android / mobile AR / image targets / App Store & Google Play deployment*
- **Immersive VR Laboratory Training System:** Developed procedural VR training workflows combining interactive tasks, UI/UX, simulation logic, systems integration, testing, and XR deployment. **Technologies:** *Unity / C# / VR interaction systems*
- **Virtual Tour of the College of Engineering:** Developed an immersive virtual-tour experience integrating interactive 3D content, navigation, and user-facing presentation workflows. **Technologies:** *Unity / interactive 3D / VR*
- **Heart of the Nation Virtual Reality Exhibition:** Supported development and delivery of an immersive VR exhibition experience requiring content integration, interaction implementation, and deployment readiness. **Technologies:** *Unity / VR / interactive media*

PROFESSIONAL EXPERIENCE

Assistant Teaching Professor / Software Development Lead — University of Missouri-Columbia 2022 – 2026

- Directed applied full-stack and systems-development work across React, Angular, Next.js, Node.js/Express, REST APIs, databases, authentication, Docker, Linux servers, and deployment workflows.
- Led software teams through requirements analysis, architecture, implementation, Git-based collaboration, code review, debugging, testing, documentation, deployment, and technical handoff.
- Designed and built production-oriented development environments and prototypes spanning web applications, server-side systems, networking, AI-assisted software, and immersive technology.
- Served as the primary technical advisor for web-development capstone projects, helping teams make architecture, API, database, security, deployment, and maintainability decisions.

Adjunct Professor – VR / Networking / Software Development - University of Missouri-Columbia 2019 – Present

- Built and supported Unity/C# applications, VR interaction systems, networked applications, game-development workflows, and hardware/software integrations with emphasis on practical implementation and troubleshooting.
- Developed technical prototypes and training environments involving networking, interactive 3D systems, device integration, UI/UX workflows, real-time interaction logic, and software deployment.
- Reviewed implementation approaches, diagnosed software and networking failures, and mentored developers on object-oriented design, version control, performance, and maintainable code.

Research Assistant / Senior Developer - College of Engineering, University of Missouri-Columbia 2017 – 2021

- Developed immersive training applications, VR simulations, mobile AR applications, and interactive systems using Unity, C#, XR hardware, cameras, networking, and real-time media integration.
- Led technical implementation for multidisciplinary projects, translating stakeholder requirements into application architecture, interaction systems, user interfaces, integration plans, and deployable builds.
- Supervised junior developers and researchers, coordinated project work, resolved cross-stack software/hardware issues, and supported testing, demonstrations, and production readiness.

EDUCATION

M.S., Computer Science — University of Missouri-Columbia

May 2022

Thesis: XR Development for Educational Purposes | Advisor: Prof. Fang Wang

B.S., Information Technology — University of Missouri-Columbia

May 2018

Minor in Computer Science | Major GPA: 3.6/4.0

TECHNICAL LEADERSHIP & RECOGNITION

- **MUVR — Founder & Vice President (2017–2018):** Founded an XR-focused technology organization, led collaborative development initiatives and workshops, and helped establish a sustainable student development community; organization received the Chancellor's Excellence Award for Best New Organization.

PROFESSIONAL REFERENCES

Fang Wang | Associate Teaching Professor; Director, CREXR Lab

Electrical Engineering and Computer Science, University of Missouri | M.S. Thesis Advisor | wangfan@missouri.edu | 573-884-8747

Hani Salim, P.E. | Professor; Chair, Engineering and Information Technology

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